

### MA.912.AR.1.4

#### **Benchmark**

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**MA.912.AR.1.4** Divide a polynomial expression by a monomial expression with rational number coefficients.

#### Benchmark Clarifications:

*Clarification 1:* Within the Algebra I course, polynomial expressions are limited to 3 or fewer terms.

#### **Connecting Benchmarks/Horizontal Alignment**

- MA.912.NSO.1.1
- MA.912.F.3.1

#### **Terms from the K-12 Glossary**

- Monomial
- Polynomial

#### **Vertical Alignment**

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##### **Previous Benchmarks**

- MA.7.AR.1.1
- MA.8.AR.1.2
- MA.8.AR.1.3

##### **Next Benchmarks**

- MA.912.AR.1.5, MA.912.AR.1.6, MA.912.AR.1.7
- MA.912.AR.6.3

## Purpose and Instructional Strategies

In middle grades, students added, subtracted and multiplied linear expressions. In Algebra I, students perform operations on polynomials limited to 3 or fewer terms and divide polynomials by a monomial. In later courses, students will perform operations on all polynomials.

- Instruction includes the connection to addition, subtraction and multiplication of polynomials to develop the understanding of closure, and the connection to properties of exponents.
- Instruction includes proper vocabulary and terminology, keeping in mind that when dividing, the word “cancel” can become a misconception for students. A number or an expression divided by itself is equivalent to 1 and does not disappear (see Appendix D).
- Instruction includes the use of manipulatives, like algebra tiles, and various strategies, like the area model, properties of exponents and decomposing fractions.

- Decomposing fractions

The expression  $\frac{12mn^6 - 40m^2n^3}{4m^2n^3}$  can be written as  $\frac{12mn^6}{4m^2n^3} - \frac{40m^2n^3}{4m^2n^3}$ . Students can then perform the division with each fraction to determine the difference.

## Common Misconceptions or Errors

- Students may not understand the meaning of closure (although not directly discussed in this benchmark, polynomials are not closed under division).
- Students may not understand like terms.

## Strategies to Support Tiered Instruction

- Teacher provides examples showing that polynomials are closed under the operations of addition, subtraction and multiplication, but not under division.
  - For example, when subtracting or multiplying polynomials (as shown below), the result is always a polynomial.

$$7x + 5 - (3x + 8) = 4x - 3$$

$$(7x + 5)(3x + 8) = 21x^2 + 71x + 40$$

- For example, when dividing polynomials (as shown below), the result may or may not be a polynomial.

$$\frac{2x^2 + 5x}{x} = 2x + 5 \text{ (which is a polynomial)}$$

$$\frac{2x^2 + 5x + 1}{x} = 2x + 5 + \frac{1}{x} \text{ (which is not a polynomial)}$$

## Instructional Tasks

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### Instructional Task 1 (MTR.3.1, MTR.4.1, MTR.5.1)

Part A. Determine the quotient of  $\left(\frac{1}{3}a^4 - 3a^3 + \frac{1}{2}a^2\right)$  and  $3a^3$ .

Part B. Discuss with your partner the strategy used. How do your quotients compare to one another?

### Instructional Task 2 (MTR.3.1, MTR.4.1, MTR.5.1)

Part A. Determine the quotient of  $x + x^2$  and  $x^{-1}$ .

Part B. What do you notice about your answer and the Laws of Exponents?

Part C. What happens when you divide  $x + x^2$  by the expression  $x^{\frac{1}{2}}$  (which is not a monomial)?

## Instructional Items

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### Instructional Item 1

What is the quotient of the expression  $\frac{15d^2 - 25d^4}{5d^3}$ ?

### Instructional Item 2

What is the quotient of the expression  $\frac{8m^2n - 5mn^2 - 20mn}{\frac{1}{4}mn}$ ?

### Instructional Item 3

What is the quotient of the expression  $\left(-\frac{5}{12}x^3y + \frac{2}{3}x^2y^2 - \frac{1}{2}x^2y\right) \div (-6x^2y)$ ?

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*\*The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*